KITCHEN THERMODYNAMICS

Experiment 16: Pure Water — Four-Sensor Thermal Gradient Inversion

Part XVI of the Kitchen Thermodynamics Series

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This paper is presented in accordance with the formal dual-format standards established by the foundational work: *Dual-Format Scientific Publishing: Optimizing Knowledge Transfer for Human and AI Cognition.*

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1 Abstract

Pure water (1 liter) in a Tramontina triple-bottom stainless steel pot is heated from ambient temperature ($\approx 26\,°\mathrm{C}$) under maximum gas flame while monitored simultaneously by four independent temperature sensors: two digital probe thermometers (t1 = fast-reading, t2 = standard, both touching the pot bottom) and two handheld infrared thermometers (IR1 and IR2, measuring the water surface). Data are recorded at 10-second intervals from $t = 0$ to $t = 150\,\mathrm{s}$.

Three anomalies are documented: (1) the pot bottom does not heat during the first 10 seconds of maximum flame — t2 registers a *decrease* of $0.1\,°\mathrm{C}$ while fire is active; (2) the water surface (IR) is consistently hotter than the pot bottom (probes) throughout the entire 150-second experiment, with $\Delta T_{\mathrm{surface-bottom}}$ ranging from $3.7\,°\mathrm{C}$ at $t = 10\,\mathrm{s}$ to $5.1\,°\mathrm{C}$ at $t = 150\,\mathrm{s}$; (3) this pattern is identical to that observed in Experiment 5 (beans & lentils) but occurs in *pure water*, eliminating organic composition as an explanatory variable.

These observations falsify two classical predictions: that the bottom of a pot heated from below is the hottest point, and that temperature rises monotonically with sustained heating from the moment heat is applied. The results are consistent with the thermal force law $\vec{F}_{\mathrm{thermal}} = -k_T\, m\, \nabla T$ established in the Kitchen Thermodynamics series.

2 Introduction

Experiments 5 and 6 of the Kitchen Thermodynamics series documented anomalous thermal behavior in water containing beans and lentils: temperature *decreased* by $1.5\,°\mathrm{C}$ during 41 seconds of maximum flame (Exp. 5), and temperature rose $+11.7\,°\mathrm{C}$ over 17 minutes *after* flame removal (Exp. 6). Both experiments also documented an inverted thermal gradient — the bottom of the pot (directly above the flame) was colder than the liquid above it.

A natural objection to these findings is that the organic matter (beans, lentils, oils, dissolved compounds) may create anomalous thermal behavior through endothermic chemical reactions or altered convection patterns specific to complex mixtures. Experiment 16 addresses this objection directly by replicating the core measurement protocol — simultaneous multi-point thermometry during heating from below — using *pure water* with no additives.

The classical predictions under test are:

- **P16-1:** The bottom of a pot heated from below is the hottest point at all times during heating.
- **P16-2:** Temperature at the bottom increases monotonically from the moment external heat is applied.
- **P16-3:** The thermal gradient points downward (hot at bottom, cool at top) when heating from below.

All three predictions are violated.

3 Experimental Protocol

Table 1: Experiment 16 — Experimental Parameters

Parameter	**Value**
Container	Tramontina triple-bottom stainless steel pot
Liquid	Pure filtered water, 1 liter
Heat source	Domestic gas flame, maximum intensity
Sensor t1	Digital fast-reading probe thermometer (bottom)
Sensor t2	Digital standard probe thermometer (bottom)
Sensor IR1	Handheld infrared thermometer (surface)
Sensor IR2	Handheld infrared thermometer (surface)
Ambient verification	Medical-grade ambient thermometer: 26.0 °C
Data interval	Every 10 seconds
Duration	$t = 0$ to $t = 150$ s (2 min 30 s)
Recording	Continuous smartphone video

Sensor placement: Both probe thermometers (t1, t2) are immersed in the water with tips touching the pot bottom — the point of maximum thermal contact with the flame below. Both IR thermometers (IR1, IR2) are aimed at the water surface from above. This configuration provides simultaneous measurement of the thermal state at the two extremes of the vertical water column: the point nearest the heat source (bottom) and the point farthest from it (surface).
Significance of four simultaneous sensors: Previous experiments in the series used one or two thermometers. Experiment 16 deploys four independent sensors to eliminate instrument-specific artifacts. The two bottom probes (t1, t2) have different response times (fast vs. standard) to test whether apparent anomalies are response-time artifacts. The two IR sensors provide redundant surface measurements.

4 Baseline Measurements (Pre-Flame)

At $t = 0$, before flame ignition, the four sensors read:

Table 2: Baseline readings at $t = 0$ (flame OFF)

Sensor	**Position**	**Reading (°C)**
Medical thermometer	Ambient air	26.0
IR1	Water surface	25.8
IR2	Water surface	25.8
t1 (fast)	Pot bottom	27.3
t2 (standard)	Pot bottom	26.3

Baseline observations: The bottom probes read slightly higher than the surface ($\Delta \approx 1\,°\mathrm{C}$), consistent with residual thermal contact with the pot material at room temperature.

IR1 and IR2 agree exactly (25.8 °C). The t1–t2 difference (1.0 °C) reflects sensor calibration offset and/or micro-positioning differences — this offset persists as a systematic bias but does not affect the gradient direction analysis.

5 Complete Data: $t = 0$ to $t = 150\,\mathrm{s}$

Table 3: Complete four-sensor data, Experiment 16. All temperatures in °C.

t (s)	Flame	t1 (fast)	t2 (std)	IR1	IR2	$\overline{\Delta}$ (IR−probe)
0	OFF	27.3	26.3	25.8	25.8	−1.0
10	ON	27.3	26.2	31.0	30.0	+3.8
20	ON	29.4	27.0	33.7	34.1	+5.7
30	ON	29.7	28.7	38.2	41.5	+10.7
40	ON	31.2	31.6	42.6	44.8	+12.3
50	ON	33.1	34.1	45.7	44.1	+11.3
60	ON	37.1	37.0	48.8	46.7	+10.7
70	ON	40.5	40.4	51.3	48.5	+9.5
80	ON	43.6	42.6	53.8	51.0	+9.3
90	ON	45.0	45.7	55.5	52.6	+8.7
100	ON	48.2	48.5	58.1	56.3	+8.9
110	ON	51.9	51.0	59.7	57.9	+7.4
120	ON	52.9	53.7	62.0	58.9	+7.2
130	ON	56.8	56.5	63.6	59.7	+5.0
140	ON	58.2	58.8	64.7	65.8	+6.8
150	ON	62.1	63.8	67.2	67.2	+4.3

$\overline{\Delta}$ is calculated as $\frac{(\mathrm{IR1}+\mathrm{IR2})}{2} - \frac{(\mathrm{t1}+\mathrm{t2})}{2}$. Positive values indicate the surface is hotter than the bottom.

6 Photographic Documentation

6.1 Collage C1 — Baseline and Sensor Setup (4 Frames)

[Collage placeholder — 4 frames from video showing baseline setup with all 4 sensors, followed by flame ignition.]

INFERENTIAL FUNCTION: The baseline collage establishes four independent instruments reading consistent temperatures at thermal equilibrium. The slight bottom-is-warmer gradient (+1 °C) before flame ignition is the classical expectation. Everything that follows documents the *inversion* of this gradient.

6.2 Collage C2 — First 40 Seconds: Gradient Inversion Onset (4 Frames)

[Collage placeholder — 4 frames at t=10, 20, 30, 40 s showing all 4 sensor displays.]

ANOMALY AT $t = 10\,\mathrm{s}$: The fast-reading probe (t1) is *unchanged* at 27.3 °C after 10 seconds of maximum flame directly beneath it. The standard probe (t2) has *decreased* from 26.3 °C to 26.2 °C. Meanwhile, the water surface has jumped from 25.8 °C to 30.5 °C (average of IR1 and IR2). The gradient has *inverted* in the first 10 seconds: the surface is now hotter than the bottom, and the bottom has not warmed at all.

6.3 Collage C3 — $t = 50$ to $80\,\mathrm{s}$: Sustained Inversion (4 Frames)

[Collage placeholder — 4 frames at t=50, 60, 70, 80 s.]

The inverted gradient persists: surface consistently +9 to +11 °C hotter than bottom. Both probes now rising (33–43 °C range), but always lagging the surface sensors.

6.4 Collage C4 — $t = 90$ to $120\,\mathrm{s}$: Gradient Convergence Begins (4 Frames)

[Collage placeholder — 4 frames at t=90, 100, 110, 120 s.]

The gap narrows from +8.7 °C to +7.2 °C. The system is beginning to equilibrate, but the inversion persists at $t = 120\,\mathrm{s}$ — two full minutes of heating.

6.5 Collage C5 — $t = 130$ to $150\,\mathrm{s}$: Final State (3 Frames)

[Collage placeholder — 3 frames at t=130, 140, 150 s.]

At $t = 150\,\mathrm{s}$, all sensors are converging: probes at 62–64 °C, IR at 67.2 °C. The surface is *still* +4.3 °C hotter than the bottom after 2 minutes 30 seconds of maximum flame. The classical prediction (bottom hottest) has not been realized at any point during the experiment.

7 Anomaly Analysis

7.1 Anomaly W16-1: Bottom Does Not Heat in First 10 Seconds

At $t = 10\,\mathrm{s}$ with maximum flame active: t1 = 27.3 °C (unchanged from baseline), t2 = 26.2 °C (*decreased* 0.1 °C from baseline of 26.3 °C). The pot bottom is in direct thermal contact with the flame via the triple-bottom stainless steel base. Classical heat transfer predicts immediate temperature increase at the contact surface.

This reproduces exactly the Experiment 5 finding (beans & lentils dropped 1.5 °C in 41 s), but in *pure water*. The organic composition hypothesis is eliminated.

Interpretation: The thermal force field $\vec{F}_{\text{thermal}} = -k_T\, m\, \nabla T$ attracts cold matter toward the heat source. The probe at the bottom measures the *incoming cold stream* — water pulled downward by the thermal field. The flame's energy is transferred upward (via torque-repulsion) before it can raise the bottom temperature. The probe at the bottom is measuring the continuously replenished cold intake, not the heated output.

7.2 Anomaly W16-2: Surface Hotter Than Bottom Throughout

The surface-bottom temperature difference $\overline{\Delta}$ is positive (surface hotter) for all 15 data points from $t = 10\,\mathrm{s}$ to $t = 150\,\mathrm{s}$. Peak inversion: $\overline{\Delta} = +12.3\,^\circ\mathrm{C}$ at $t = 40\,\mathrm{s}$. The gradient never reverses to the classical configuration (bottom hotter) during the entire experiment.

Classical convection theory: heat is applied at the bottom $\rightarrow$ bottom water heats first $\rightarrow$ hot water rises by buoyancy $\rightarrow$ cold water descends to be heated $\rightarrow$ at all times, the bottom layer is the hottest because it receives heat first. This prediction fails from $t = 10\,\mathrm{s}$ onward.

7.3 Anomaly W16-3: Pattern Identical to Beans & Lentils (Exp. 5)

Experiment 5 documented: (a) temperature decrease at the bottom during active heating; (b) inverted gradient (bottom colder than above); (c) motion beginning while temperature drops. Experiment 16 reproduces (a) and (b) in pure water. The compositional variable (organic matter, dissolved compounds, oil films) is controlled away. The anomaly is a property of *water under active thermal flux*, not of the specific solute.

8 Quantitative Analysis

8.1 Gradient Inversion Magnitude

Average surface temperature: $\bar{T}_{\mathrm{IR}} = \frac{\mathrm{IR1+IR2}}{2}$. Average bottom temperature: $\bar{T}_{\mathrm{probe}} = \frac{\mathrm{t1+t2}}{2}$.

At peak inversion ($t = 40\,\mathrm{s}$):

$$\overline{\Delta}_{\max} = \bar{T}_{\mathrm{IR}} - \bar{T}_{\mathrm{probe}} = \frac{42.6 + 44.8}{2} - \frac{31.2 + 31.6}{2} = 43.7 - 31.4 = +12.3\,^\circ\mathrm{C} \tag{1}$$

8.2 Heating Rate Comparison: Surface vs. Bottom

Average heating rate for surface (IR average, $t = 10$ to $150\,\mathrm{s}$):

$$\dot{T}_{\mathrm{surface}} = \frac{67.2 - 30.5}{140} \approx 0.26\,^\circ\mathrm{C/s} \tag{2}$$

Average heating rate for bottom (probe average, $t = 10$ to $150\,\mathrm{s}$):

$$\dot{T}_{\mathrm{bottom}} = \frac{63.0 - 26.75}{140} \approx 0.26\,^\circ\mathrm{C/s} \tag{3}$$

The *rates* are comparable, but the surface has a persistent *offset* of +3.8 to +12.3 °C. The surface heats *first*, and the bottom catches up over time but never overtakes it within the observation window.

8.3 Probe Cross-Validation

The two bottom probes (t1 fast, t2 standard) track each other within $\pm 1.5\,^\circ\mathrm{C}$ throughout, with the difference decreasing over time as both sensors equilibrate. At $t = 150\,\mathrm{s}$: t1 = 62.1 °C, t2 = 63.8 °C ($\Delta = 1.7\,^\circ\mathrm{C}$). This confirms the anomaly is not a single-instrument artifact.

The two IR sensors track each other within $\pm 3.3\,^\circ\mathrm{C}$ (maximum difference at $t = 30\,\mathrm{s}$), converging to exact agreement (67.2 °C) at $t = 150\,\mathrm{s}$. IR measurement variability reflects the spatial heterogeneity of the water surface temperature field.

9 Systematic Falsification Table

Table 4: Systematic falsification — Experiment 16

Classical Prediction	**Observation**	**Status**
P16-1: Bottom of pot is hottest point when heated from below	Surface is +3.8 to +12.3 °C hotter than bottom throughout entire experiment	**FALSIFIED**
P16-2: Temperature at bottom rises monotonically from moment of flame ignition	t1 unchanged, t2 *decreases* 0.1 °C in first 10 s of max flame	**FALSIFIED**
P16-3: Gradient points downward (bottom hot, top cool) when heating from below	Gradient inverted from $t = 10$ s onward; surface leads bottom for 150 s	**FALSIFIED**
Inverted gradient is caused by organic composition (Exp. 5 objection)	Pure water, no additives, same pattern	**ELIMINATED**
Thermal force law $\vec{F} = -k_T m \nabla T$: cold matter attracted to heat source; heated matter ejected to surface	Bottom cold (incoming stream), surface hot (ejected stream), persistent inversion	**CONSISTENT**

10 Connection to the Kitchen Thermodynamics Series

Experiment 16 serves as a *control experiment* for Experiments 5 and 6. The key finding — inverted thermal gradient with bottom colder than surface during heating from below — was first observed in bean-and-lentil water. The natural skeptical response ("organic compounds cause anomalous convection") is addressed by reproducing the core phenomenon in pure water.

This extends the series' universal principle: *bulk mechanical and thermal behavior is determined by the presence of active thermal flux (flame ON/OFF), not by the absolute temperature of the material or its composition.* The surface is hotter not because of some compositional effect, but because the thermal force field ejects heated matter upward while attracting cold matter downward toward the heat source.

The four-sensor protocol also provides the most robust instrumentation in the series to date, with redundant measurement at both spatial extremes and cross-validation between fast and standard response-time probes.

11 Conclusion

Experiment 16 documents a persistent inverted thermal gradient in *pure water* heated from below: the surface is consistently +3.8 to +12.3 °C hotter than the pot bottom over 150 seconds of continuous maximum flame. The bottom does not heat during the first 10 seconds — one probe registers a decrease. These observations reproduce the anomalies of Experiments 5 and 6 while eliminating organic composition as an explanatory variable. The thermal force law $\vec{F}_{\text{thermal}} = -k_T\, m\, \nabla T$ provides the unified explanation: cold matter is attracted toward the heat source (bottom reads cold), while heated matter is ejected to the surface (surface reads hot).

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Supplementary Materials

Video: https://vt.tiktok.com/ZSuWbB6Vk/
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